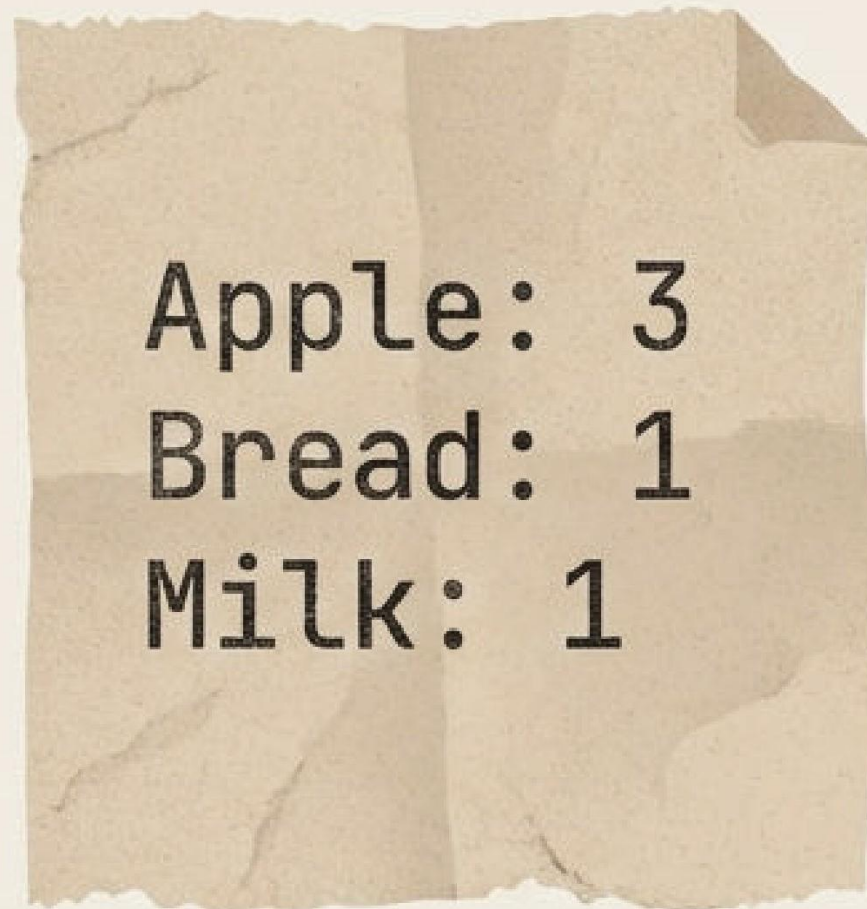


FROM COUNTING WORDS TO REASONING WITH ALIENS

A Pedagogical Evolution of Machine Understanding

PRIMITIVE PAST



SOPHISTICATED PRESENT

AI ASSISTANT

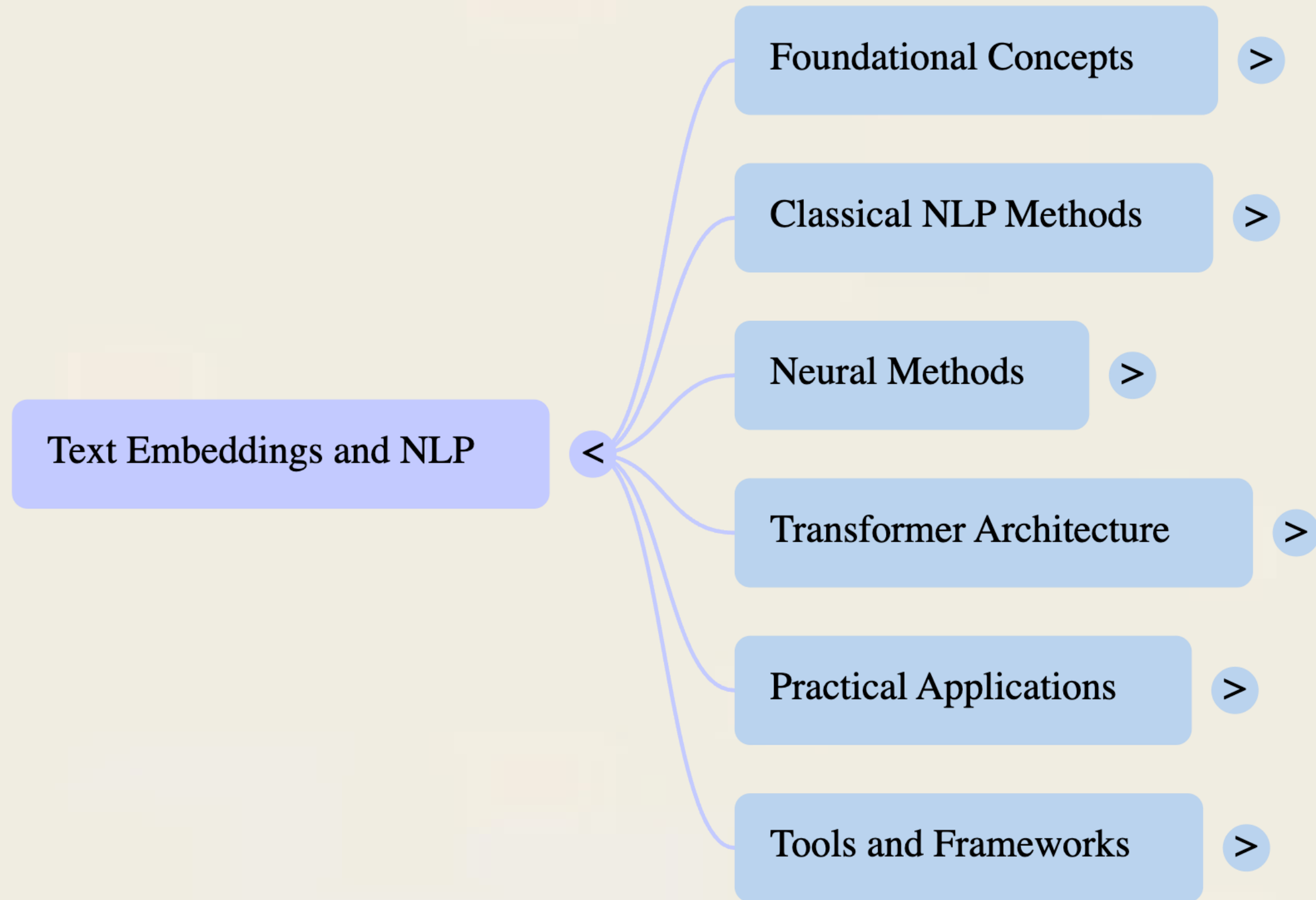


Explain relativity

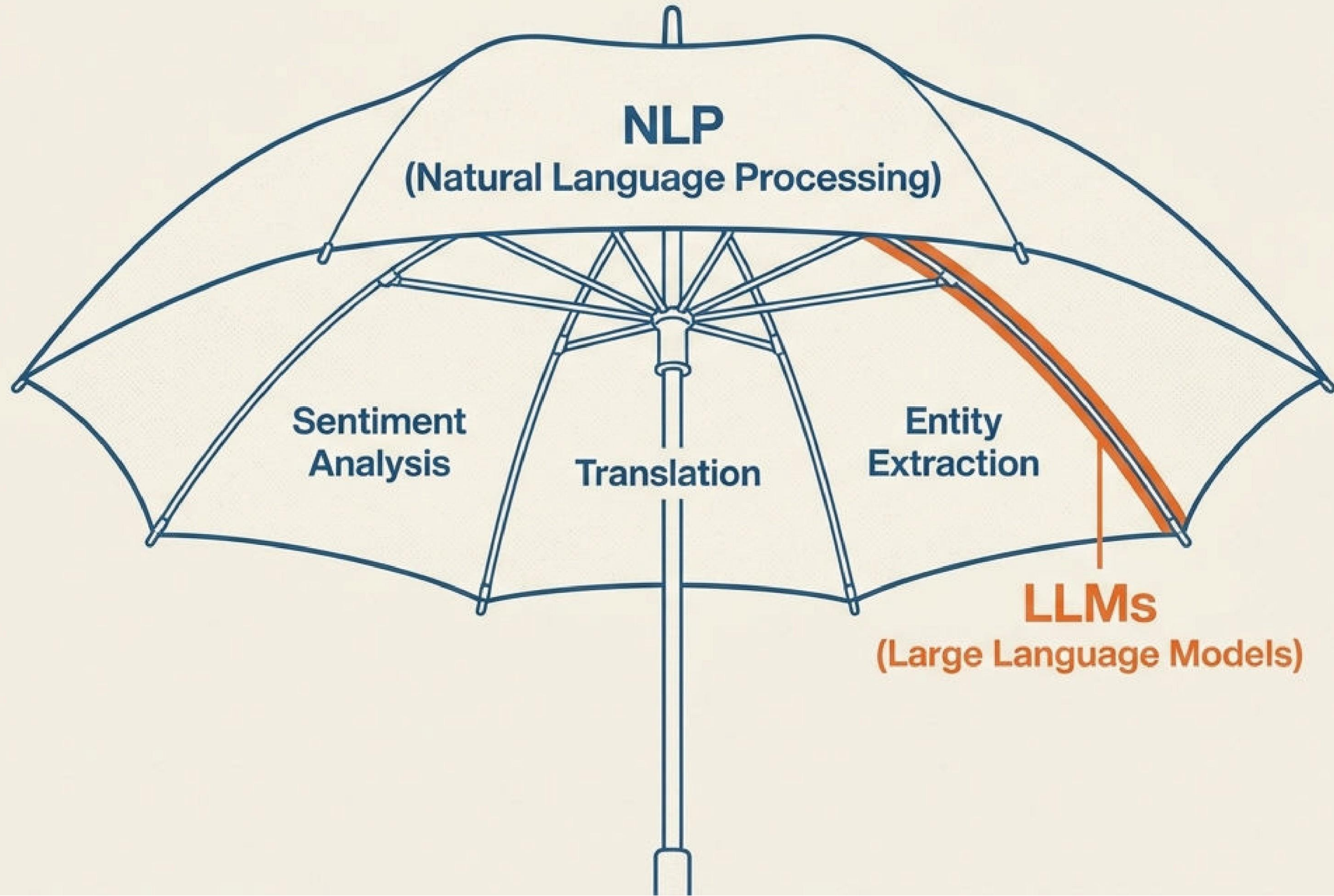


Time is relative to the observer, meaning the rate at which time passes depends on your frame of reference. This revolutionary concept, proposed by Einstein, fundamentally changed our understanding of the universe by linking space and time into a single continuum...

Computers speak numbers. Humans speak nuance. This is the story of how we taught machines to bridge the gap, climbing from simple word counts to complex reasoning.



The Fundamental Disconnect: Science vs. Tool



The Discipline (NLP)

The broad science of teaching computers to interpret human language. The 'Physics' behind the engineering.

The Tool (LLMs)

A specific, powerful engine built using NLP principles. The 'Rocket' built with physics.

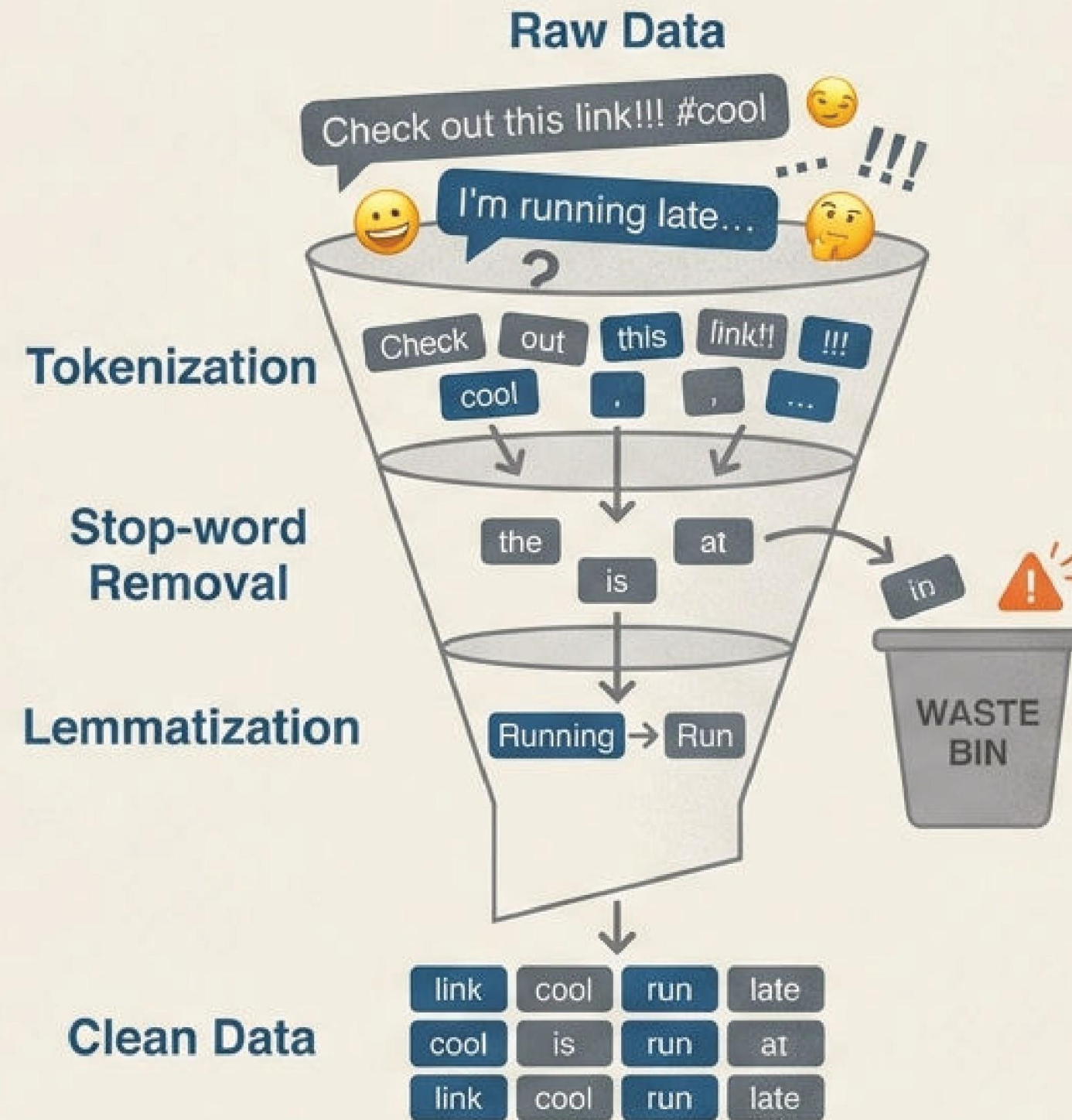
Level 1: The “Grocery List” Approach (Bag of Words)



Dog bites Man → [1, 1, 1]

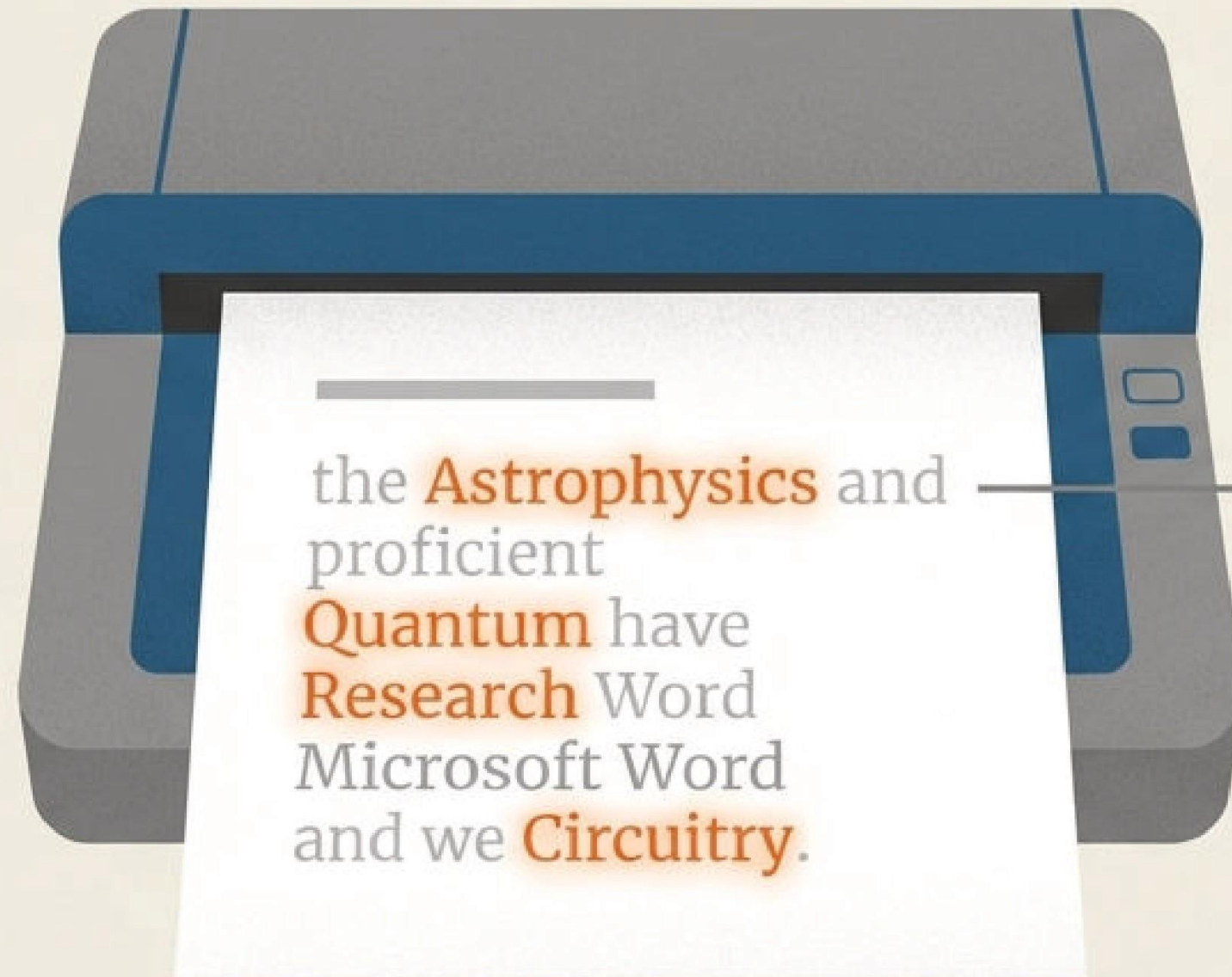
Man bites Dog → [1, 1, 1]

The Workflow: Preprocessing (Cleaning the Lens)



Garbage In, Garbage Out. We must cook the food before the AI eats.

Level 2: Weighing Importance (TF-IDF)



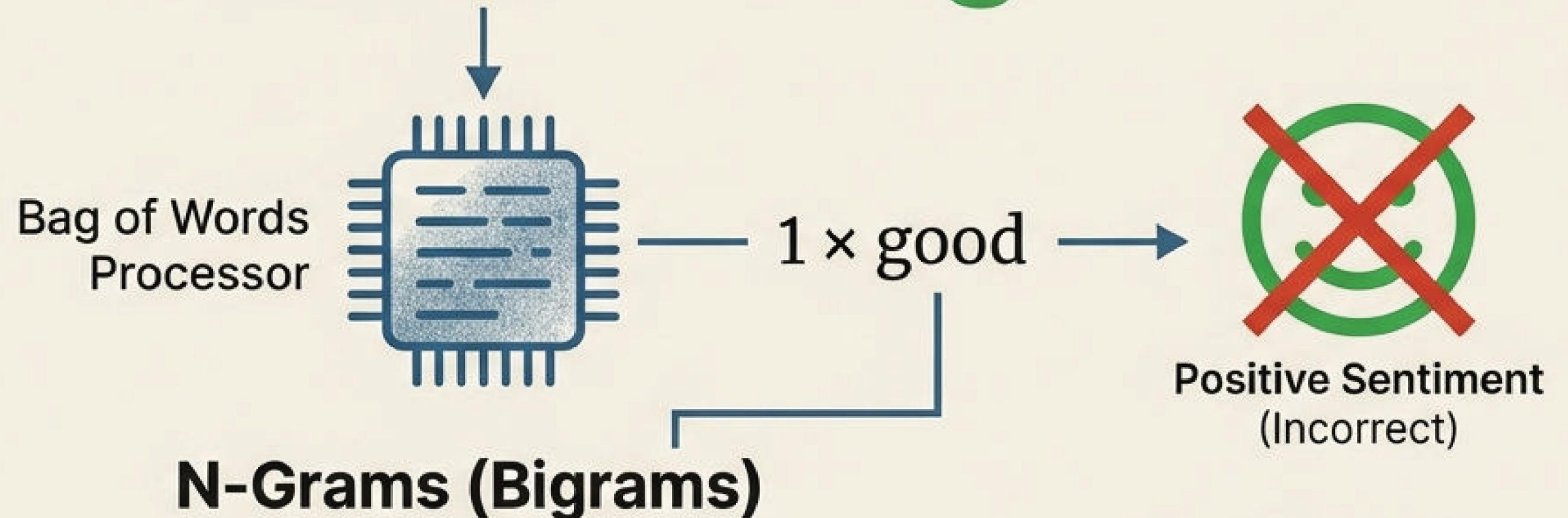
TF (Term Frequency) = How often a word appears here.
(Signal Strength)

IDF (Inverse Document Frequency) =
How rare the word is everywhere else.
(Noise Reduction)

If everyone says "Microsoft Word", it has zero value. If only you say "Astrophysics", it becomes the fingerprint of the document.

The “Context Gap”: Why Counting Fails

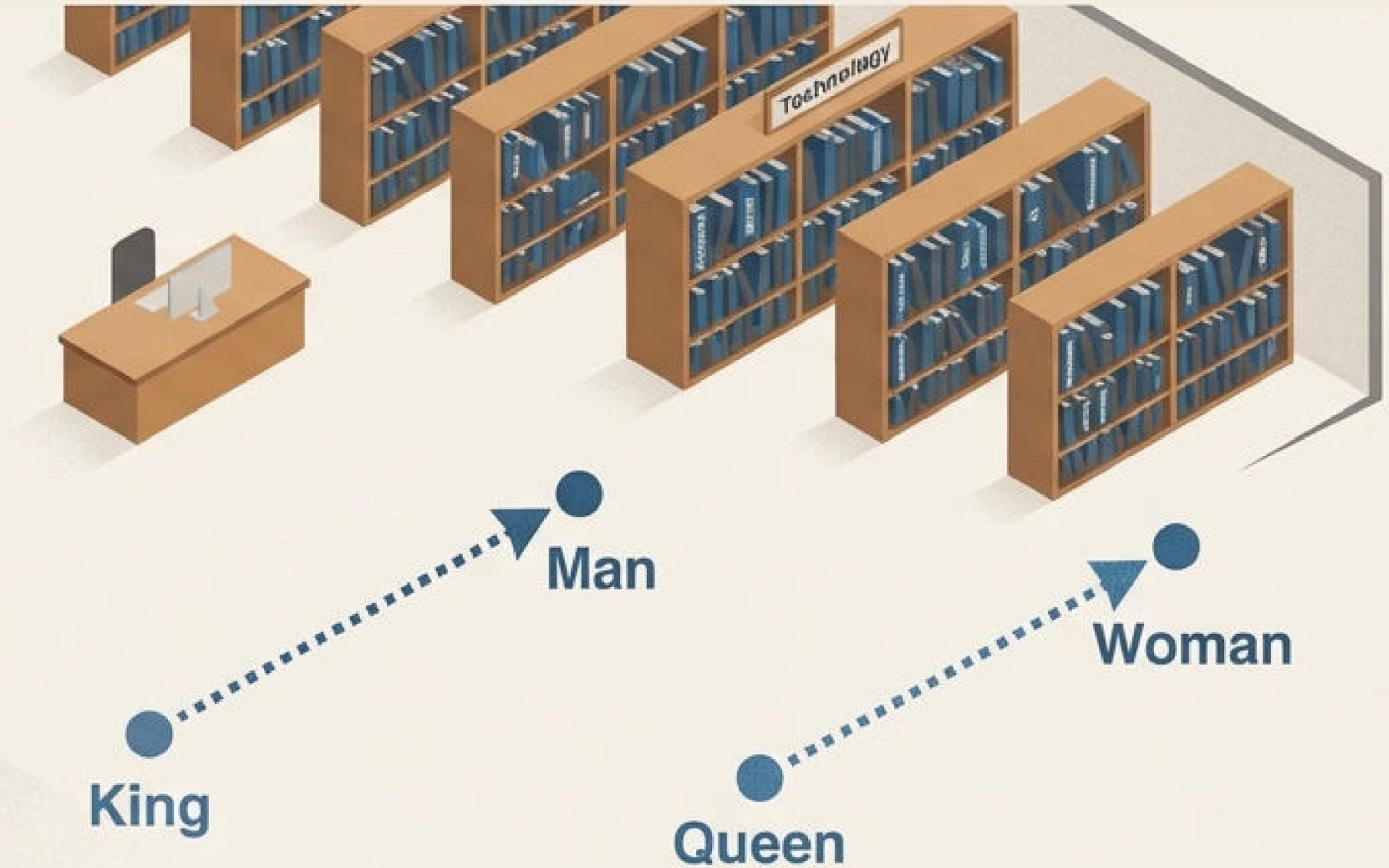
This movie is **NOT** good



["This movie"], ["movie is"], ["is not"], ["not good"] → 😞 Negative Sentiment (Correct)

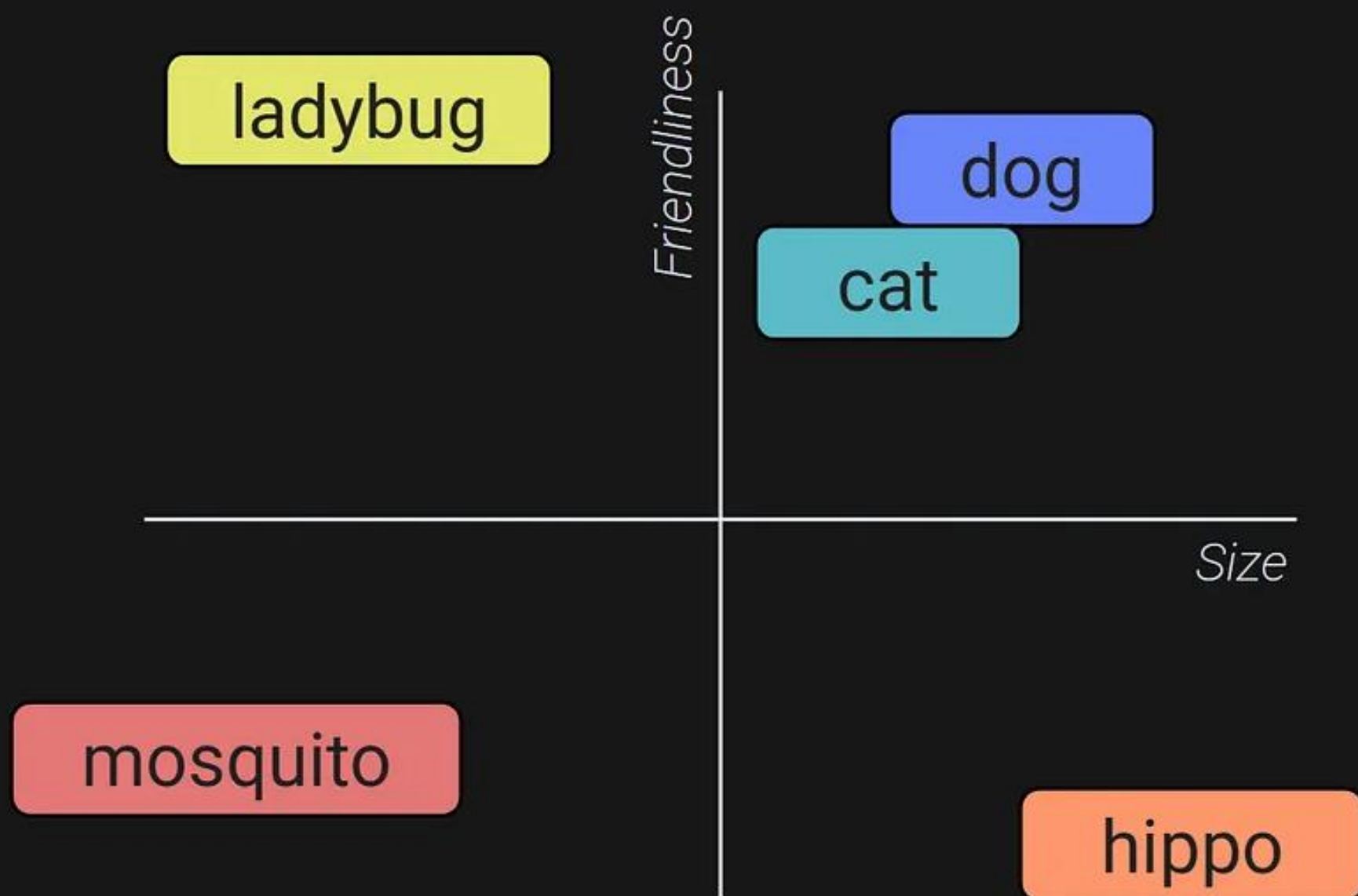
The Curse of Dimensionality: Tracking every 2-word or 3-word combination creates millions of possibilities. We need a map, not a longer list.

Level 3: The 'Map of Meaning' (Word Embeddings)



$$\text{King} - \text{Man} + \text{Woman} = \text{Queen}$$

The model understands gender not by definition, but by geometry (direction and distance).



$\text{dog} = [\quad .5, .9 \quad]$
 $\text{cat} = [\quad .4, .7 \quad]$
 $\text{ladybug} = [\quad -.7, .9 \quad]$
 $\text{hippo} = [\quad 1, -1 \quad]$
 $\text{mosquito} = [\quad -.9, -.9 \quad]$

Figure 1: Illustration of a human-crafted 2D embeddings space with five word embeddings shown. Taken from my longer [post on LLM](#) embeddings, which itself is taken from my video-based [course on LLM](#) mechanisms.

3D Coordinate System with 4 Vectors

- **Hypothetical Vectors:**

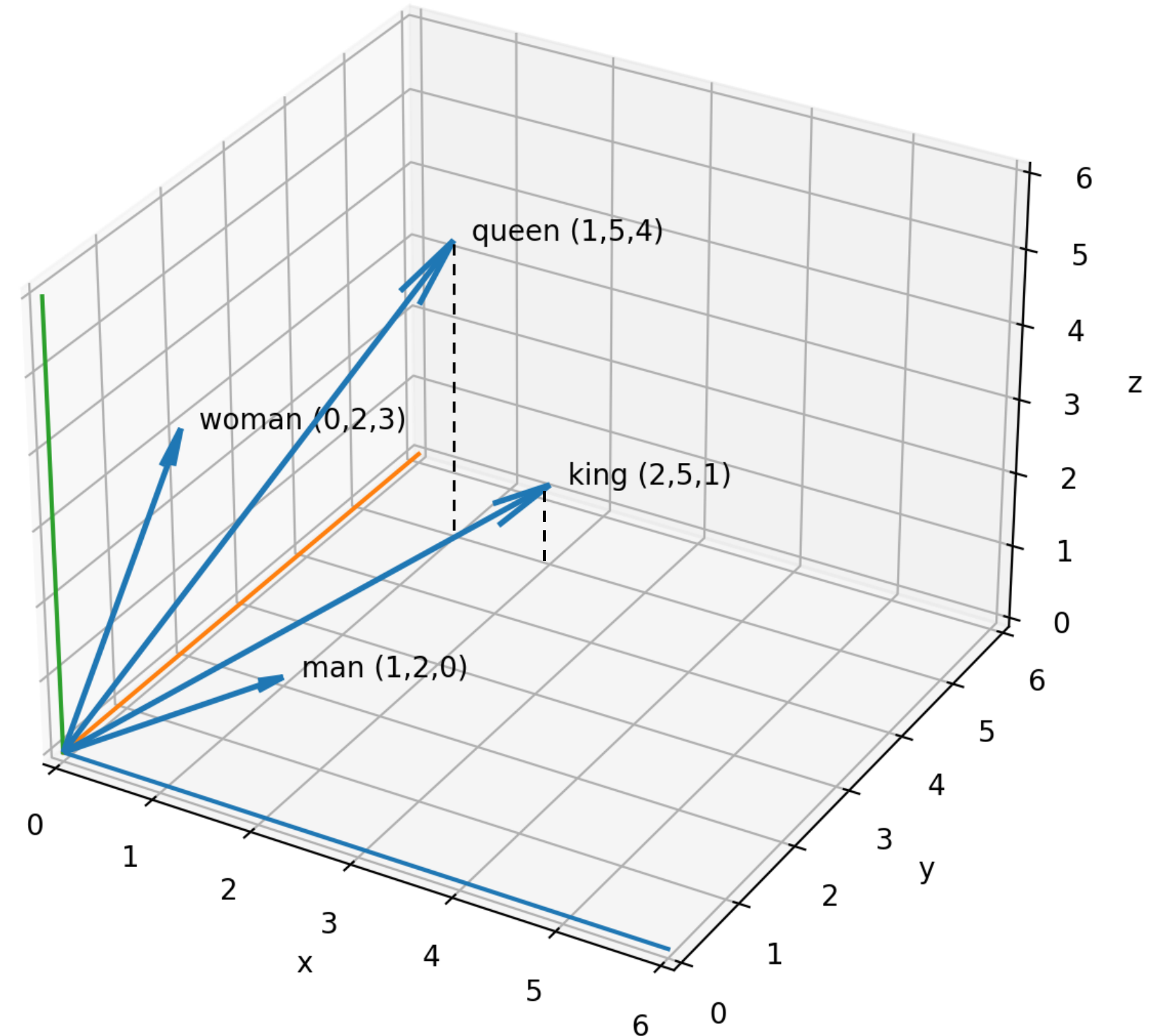
- $\vec{v}_{king} = [2, 5, 1]$
- $\vec{v}_{man} = [1, 2, 0]$
- $\vec{v}_{woman} = [0, 2, 3]$
- $\vec{v}_{queen} = [1, 5, 4]$

- **Equations:**

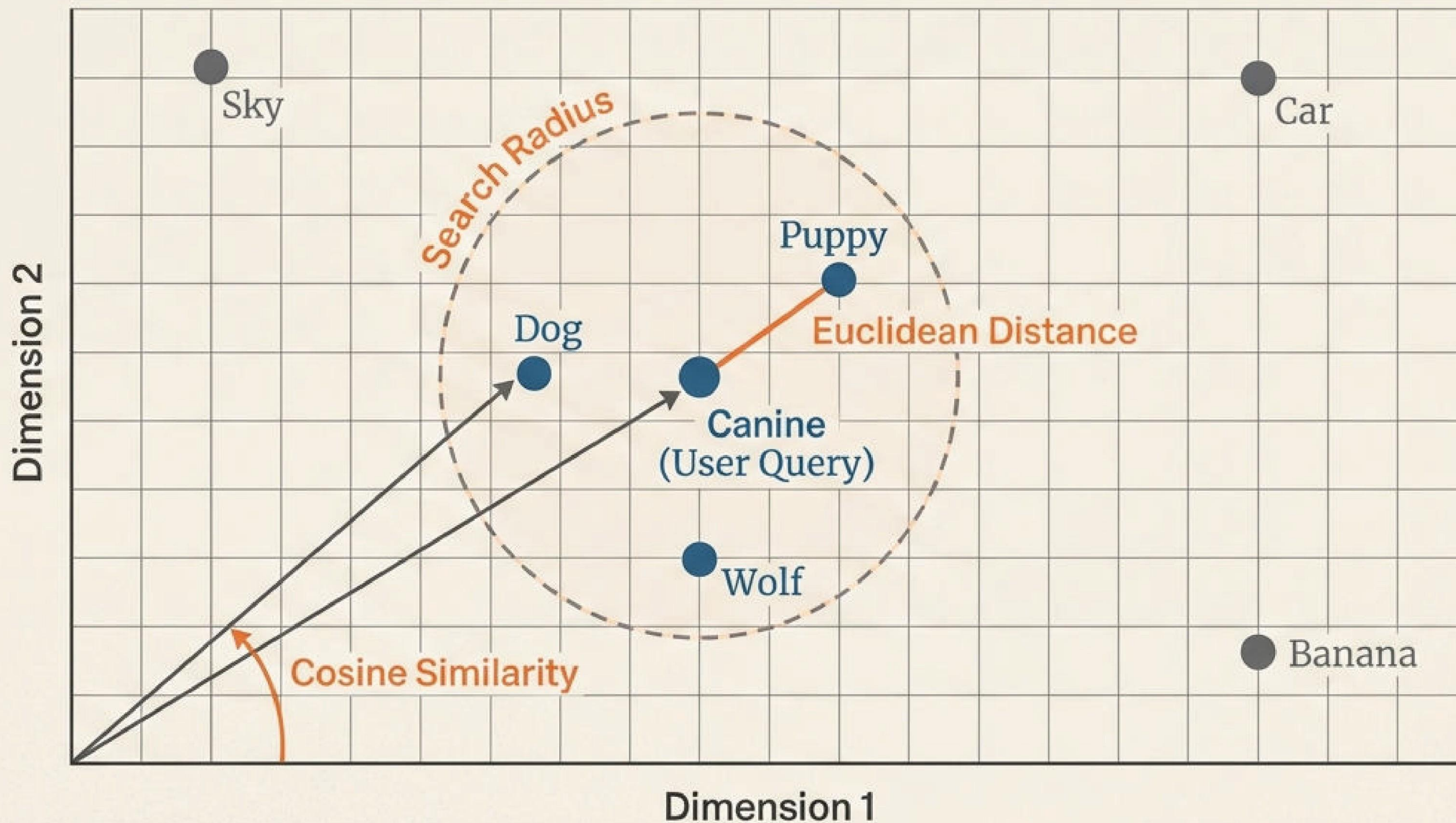
- Difference: $\vec{v}_{king} - \vec{v}_{man} = [2, 5, 1] - [1, 2, 0] = [1, 3, 1]$
- Addition: $[1, 3, 1] + \vec{v}_{woman} = [1, 3, 1] + [0, 2, 3] = [1, 5, 4]$

- **Result:**

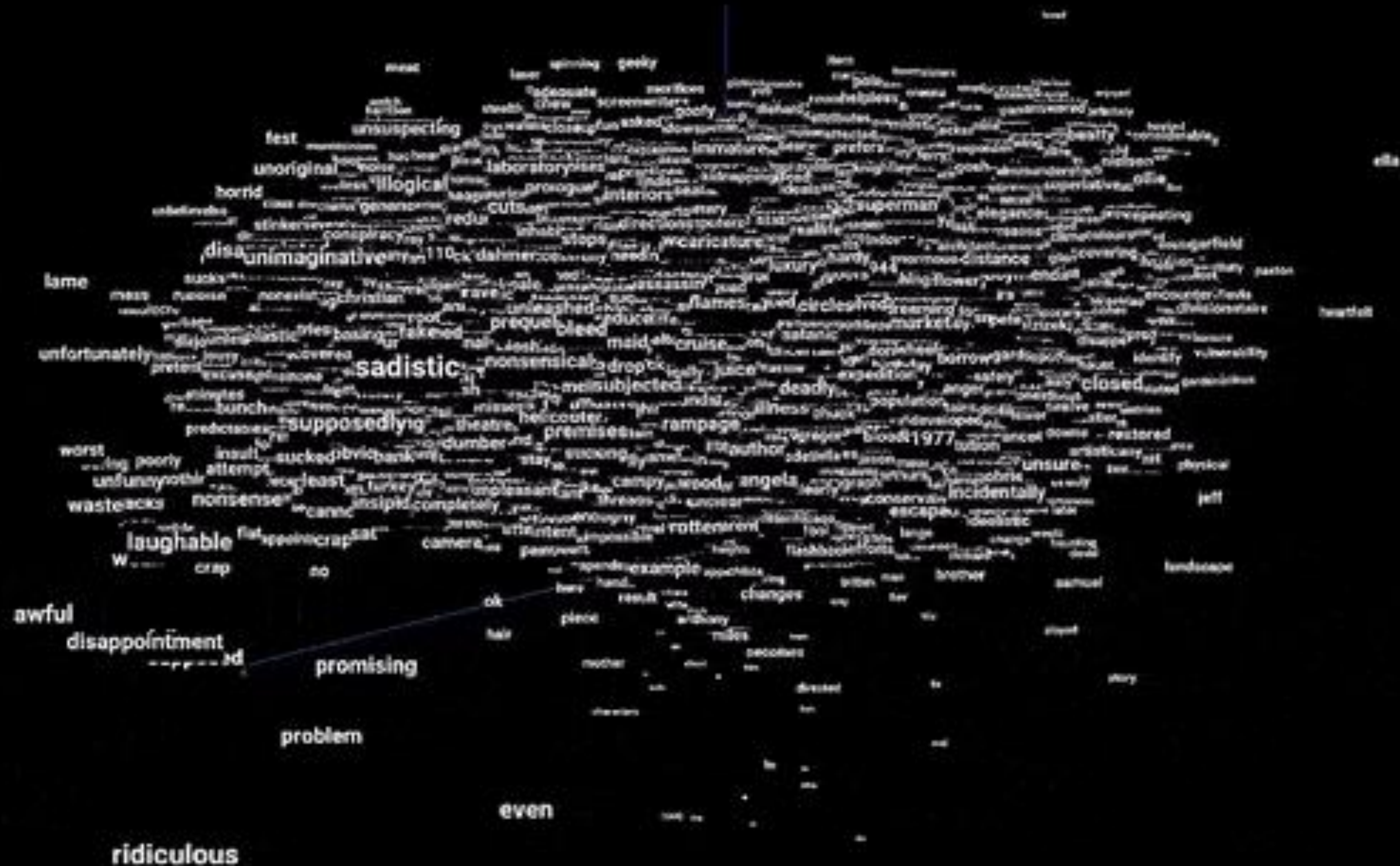
- $\vec{v}_{queen} = [1, 5, 4]$



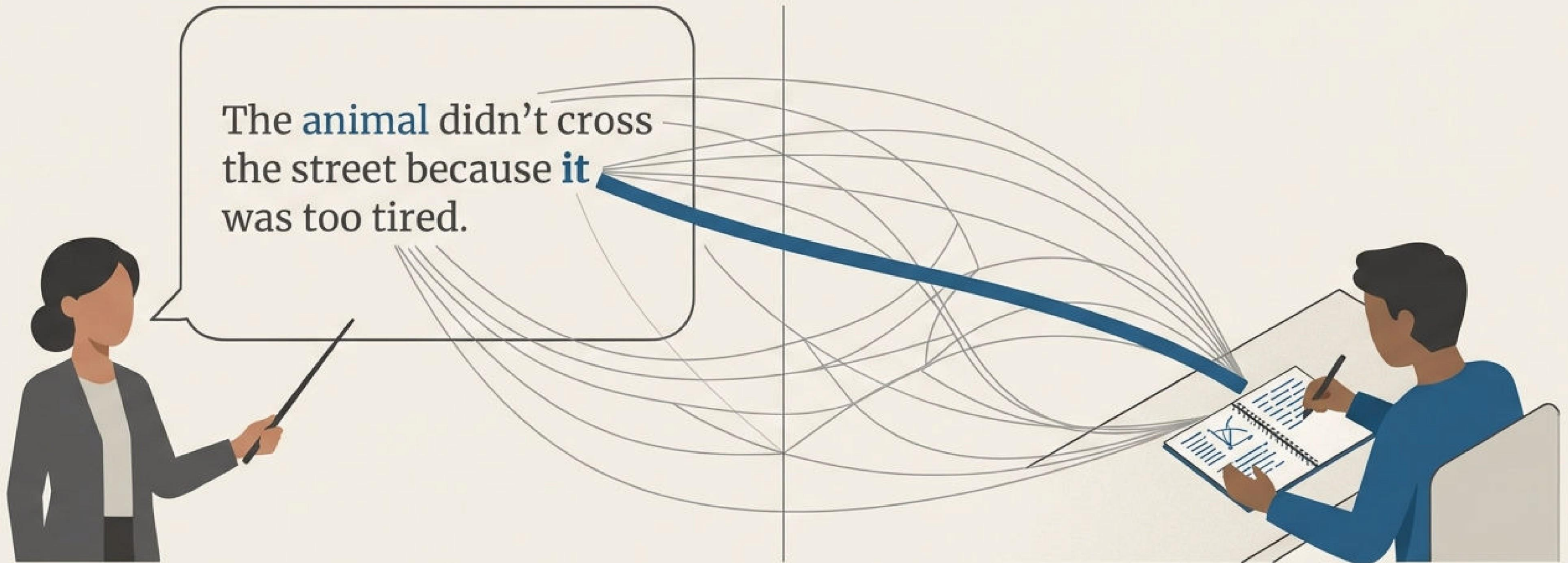
How Embeddings Work: Distance Equals Similarity



The machine doesn't know a puppy is a dog. It just knows they live in the same neighborhood of the 1,536-dimensional map.

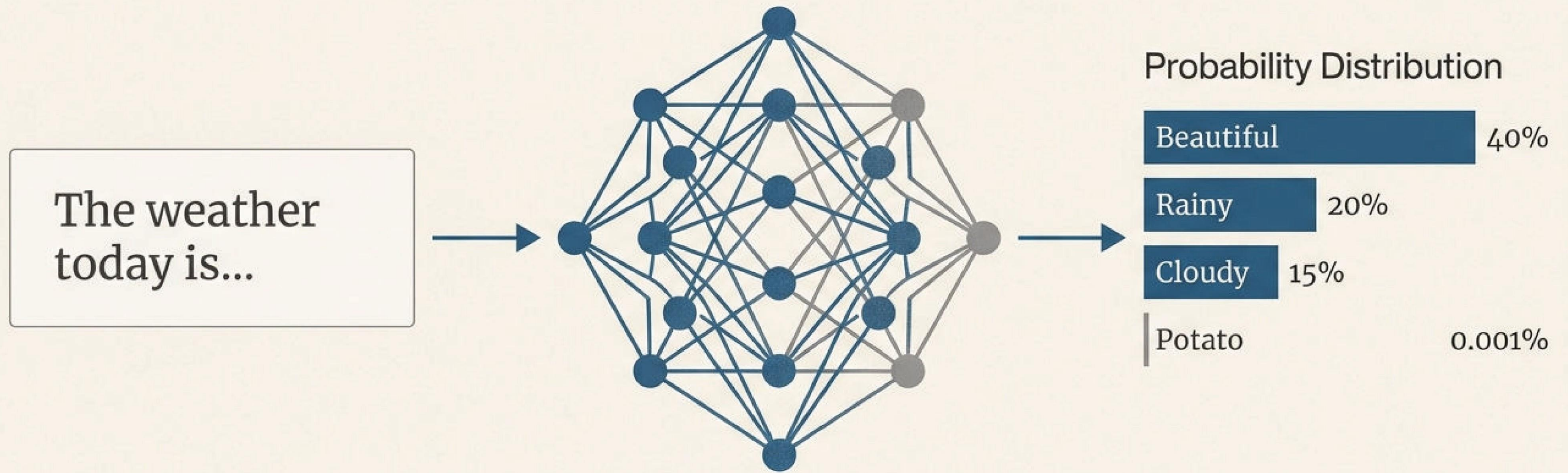


Level 4: The Revolution (Transformers & Attention)



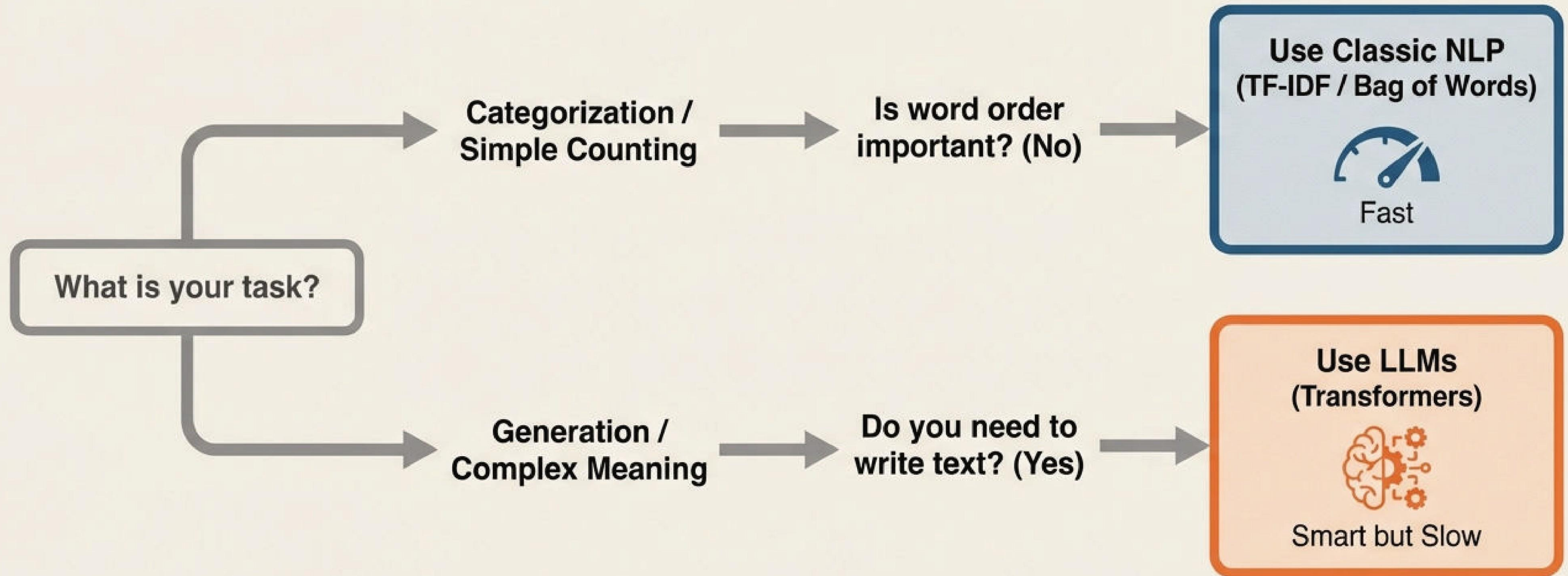
Self-Attention allows the model to dynamically understand context.
It resolves ambiguity by weighing relationships between words.

Large Language Models: Scaling the Brain



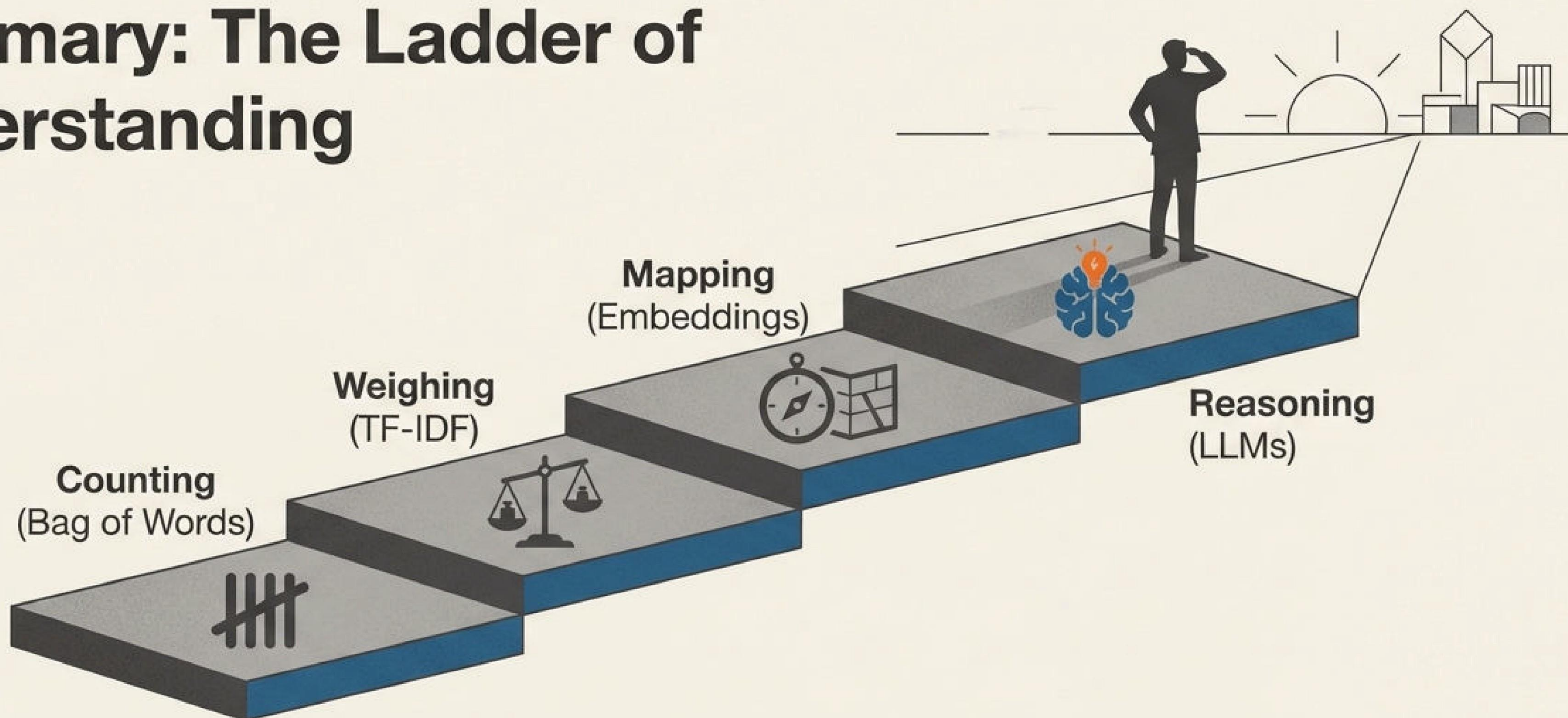
Generative AI doesn't just look up answers. It calculates the probability of the next word based on everything it has ever read.

Choosing the Right Tool



Don't use a flamethrower to light a candle. Use classic NLP for speed and transparency; use LLMs for creativity and nuance.

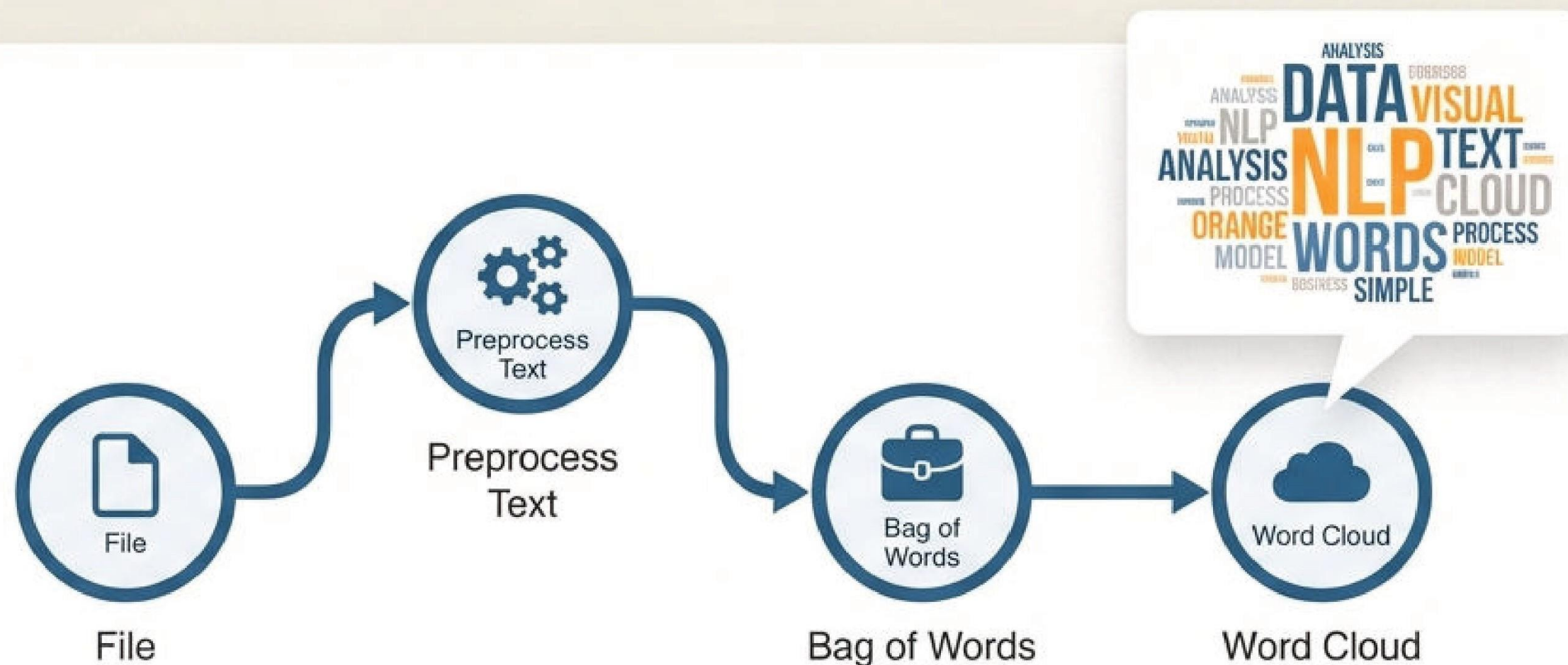
Summary: The Ladder of Understanding



We are moving from building models to **orchestrating intelligence**.
The best way to understand the aliens is to **start talking to them**.

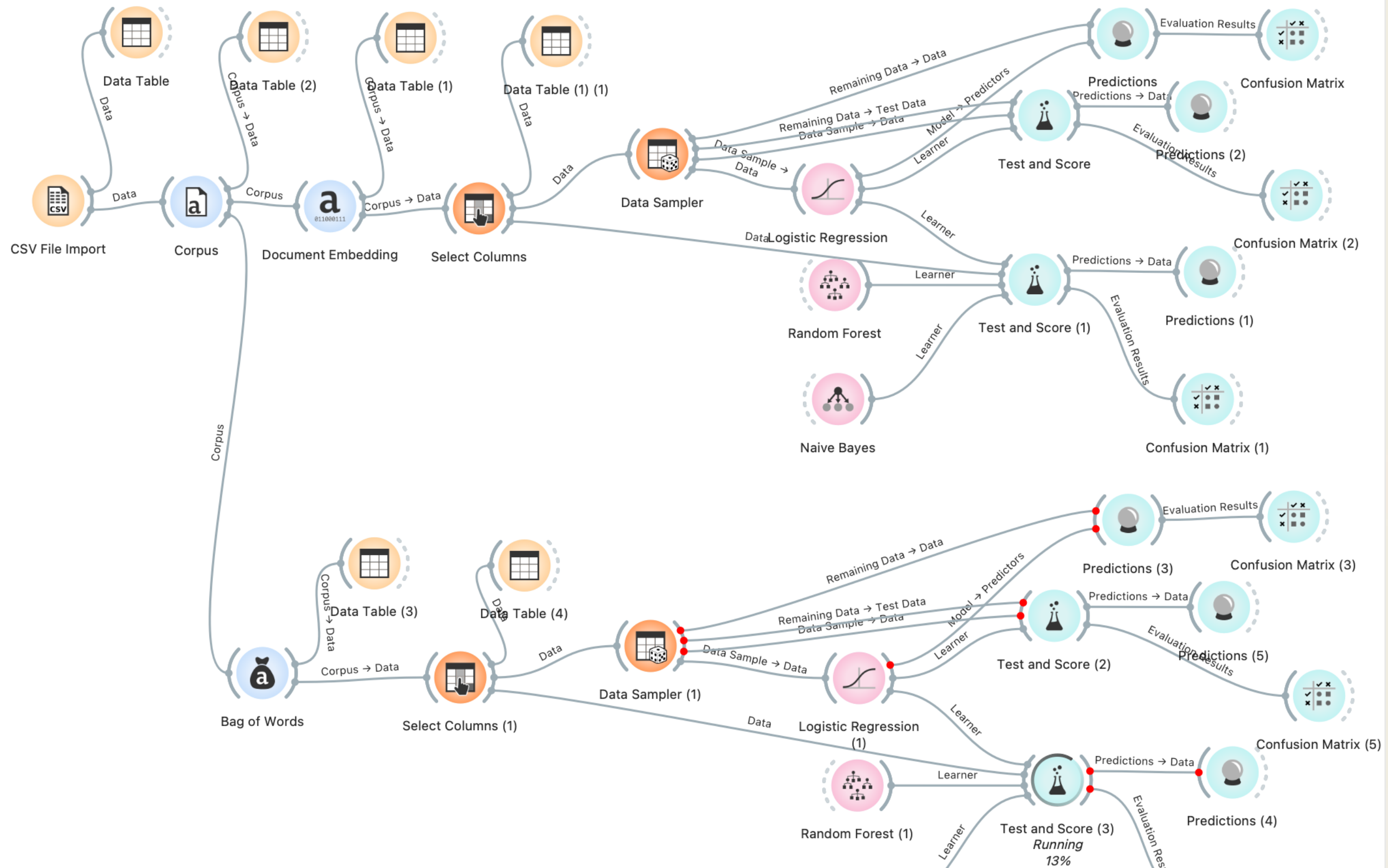
Workshop

The Toolkit: Visual Programming with Orange



Low-Code / No-Code. Drag and drop widgets to build complex NLP pipelines without writing a single line of Python.

W4 Text Classification - News Catrgory v002.ows



The Toolkit: Industry Standard (Python & Colab)

```
import openai

response = openai.Completion.create(
    engine='text-davinci-003',
    prompt='Explain relativity to a 5-year-old.'
)

print(response)
```

The Library

The Input

The Output

Google Colab provides free GPU power. Integrating with state-of-the-art AI takes less than 5 lines of code.